Experiment No:5

Course Outcome: LO5 **Blooms Level: L5**

Aim: To develop a structure-based social media analytics model for a business using network-based techniques such as community detection and influence analysis.

Abstract:

This experiment focuses on analyzing the structural properties of social media networks to extract meaningful business insights. Using graph-based modeling techniques, relationships among users are represented as networks consisting of nodes and edges. The NetworkX Python library is employed to construct, visualize, and analyze these networks. Key structural measures such as degree centrality, closeness centrality, betweenness centrality, and eigenvector centrality are computed to identify influential users, communities, and interaction patterns. Such structure-based analytics help businesses understand information diffusion, community behavior, and influencer impact in social media platforms.

Sample Input and Output:

Case 1: Create a graph representing social media users and their interactions.
Output: Nodes and edges representing users and connections are successfully created.
Case 2: Compute centrality measures for the network.
Output: Centrality values for each node are calculated.
Case 3: Visualize the social network graph.
Output: Network graph is displayed showing relationships among users.

Theory:

Structure-based social media analytics focuses on analyzing the relationships and interactions among users rather than only the content. Social media platforms naturally form complex networks where users are nodes and interactions such as likes, comments, mentions, or follows form edges. Analyzing these structures helps identify influential users, tightly connected communities, and information flow paths.

NetworkX API in Python:

NetworkX (NX) is a powerful Python library used for the creation, manipulation, and analysis of complex networks. It supports graph theory algorithms, network visualization, and structural analysis. Common operations using NetworkX include:

- Creating graphs
- Adding nodes and edges
- Visualizing networks
- Computing centrality measures

Centrality Measures:

Centrality measures quantify the importance of nodes in a network and are widely used in influence analysis.

1. Degree Centrality:

Measures the number of direct connections a node has. Nodes with high degree centrality are highly active or popular users.

2. Closeness Centrality:

Measures how close a node is to all other nodes in the network. It reflects how quickly information can spread from a node.

3. Betweenness Centrality:
Measures how often a node lies on the shortest path between other nodes. Nodes with high betweenness act as bridges or information brokers.
4. Eigenvector Centrality:
Measures a node's influence based on the importance of its neighbours. Connections to influential nodes increase the score.

Program:

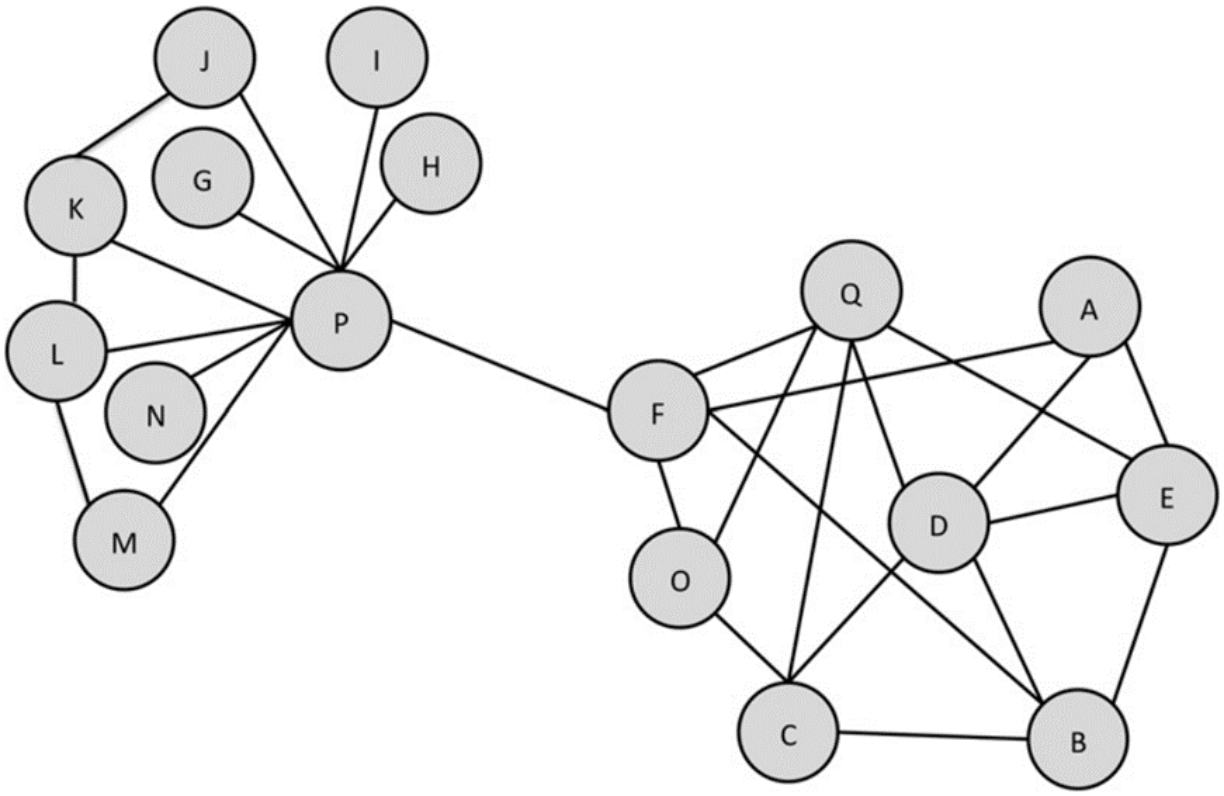
Step 1: Import required libraries
```
import networkx as nx
import matplotlib.pyplot as plt
```
Step 2: Create a graph
```
G = nx.Graph()
```
Step 3: Add nodes representing users
```
G.add_nodes_from(['A', 'B', 'C', 'D', 'E'])
```
Step 4: Add edges representing interactions
```
G.add_edge('A', 'B')
G.add_edge('A', 'C')
G.add_edge('B', 'D')
G.add_edge('C', 'D')
G.add_edge('D', 'E')
```
Step 5: Compute centrality measures
```
degree_centrality = nx.degree_centrality(G)
betweenness_centrality = nx.betweenness_centrality(G)
closeness_centrality = nx.closeness_centrality(G)
eigenvector_centrality = nx.eigenvector_centrality(G)
```
Step 6: Visualize the network
```
nx.draw(G, with_labels=True)
plt.show()
```

Output:
- Social media interaction network is successfully created.
- Degree, betweenness, closeness, and eigenvector centrality values are computed.
- Network graph visualization clearly shows user relationships and influential nodes.

Conclusion:
In this experiment, a structure-based social media analytics model was successfully developed using the NetworkX API. By modeling social media interactions as a network, various centrality measures were applied to identify influential users and key connection points. Structure-based analytics provides valuable insights into community behavior, influence patterns, and information flow, making it highly useful for business decision-making and social media strategy development.

Exercise 1:
Construct a social media interaction network for a brand's online community and identify the top three influential users using degree centrality and eigenvector centrality.

Abstract:
This exercise focuses on identifying influential users within a brand's social media network using basic centrality measures. By analyzing degree and eigenvector centrality, users with high activity and influence are identified. The outcome of this exercise helps businesses recognize key users who can effectively amplify marketing messages and improve customer engagement.



Students shall draw flowchart of exercise question in the writeup and submit.

Exercise 2:
Apply community detection techniques on a social media network graph to identify distinct user groups and analyze their interaction patterns and connectivity.

Abstract:
This exercise aims to identify communities within a social media network by applying community detection algorithms. The detected groups are analyzed to understand user interaction patterns and communication behavior. Such analysis enables businesses to segment users effectively and design targeted marketing and engagement strategies for different user communities.